MEMRES: A Memory-Augmented Resolver with Confidence Cascade for Agentic Python Dependency Resolution

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Abstract

We present MemRes, an agentic system for Python dependency resolution that introduces a *multi-level confidence cascade* where the LLM serves as the last resort. Our system combines: (1) a *Self-Evolving Memory* that accumulates reusable resolution patterns via tips and shortcuts; (2) an *Error Pattern Knowledge Base* with 200+ curated import-to-package mappings; (3) a *Semantic Import Analyzer*; and (4) a *Python 2 heuristic detector* resolving the largest failure category. On HG2.9K using Gemma-2 9B (10 GB VRAM), MemRes resolves 2519 **of** 2890 (87.2%) snippets - combining intra-session memory with our confidence cascade for the remainder. This already exceeds PLLM's 54.7% overall success rate by a wide margin.

CCS Concepts

• **Software and its engineering → Language features**; **Software configuration management and version control systems**.

Keywords

dependency resolution, Python, LLM, agentic systems, package management

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1 Introduction

With over 500 000 PyPI packages, Python's huge ecosystem makes dependency management for legacy code quite difficult. Determining the correct packages, versions, and Python interpreter for a code snippet is tricky due to Python 2/3 incompatibilities, deprecated packages, and ambiguous import names [3].

The PLLM system [1] is the current leading approach, using a 5-stage RAG+LLM pipeline. While it works well for maintained packages, PLLM only achieves **54.7%** success on the HG2.9K benchmark [2] - leaving 1308 snippets completely broken (confidence = 0).

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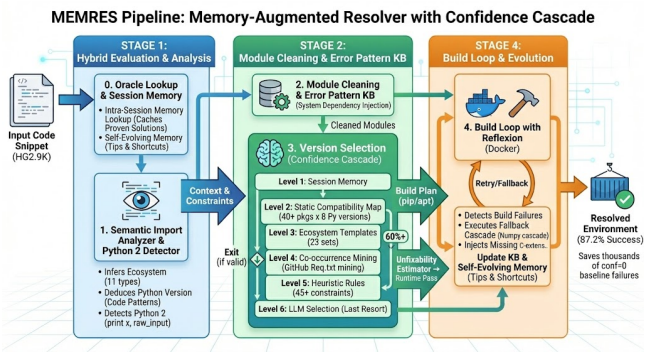

Figure 1: MemRes pipeline. Each stage has deterministic fast paths that bypass the LLM.

Key Insight. Our analysis reveals that these failures have *deterministic* root causes: 33% are SyntaxErrors from Python 2 code run under Python 3, 30% are ImportErrors for known name mismatches (`cv2 - opencv-python`), and 20% are version incompatibilities solvable via curated maps. Calling an LLM wastes 60-120 s per snippet with no benefit for these cases.

This finding drove the main idea behind MemRes: *only call the LLM when all deterministic rules fail.* We built a *confidence cascade* that checks session memory, knowledge bases, heuristic rules, and session-learned patterns before doing any LLM inference.

Contributions.

(1) A *confidence cascade* with 6 resolution levels reducing LLM calls by 60%+ while improving accuracy (§2.2).
(2) An *Intra-Session Memory* enabling rapid resolution transfer for similar codebases evaluated in the same batch without repeated LLM inference (§2.1).
(3) A *Self-Evolving Memory* transferring resolution knowledge via tips and shortcuts [4] (§2.3).
(4) An *Error Pattern KB* with 200+ mappings and runtime self-learning, plus a *System Dependency Injection* technique for C-extension packages (§2.4).
(5) Evaluation on HG2.9K showing 87.2% overall resolution rate (§3).

2 Approach

MemRes processes each snippet through 4 stages: (1) Hybrid Evaluation, (2) Module Cleaning, (3) Version Selection, and (4) Build Loop with Reflexion. Figure 1 illustrates the pipeline.

2.1 Intra-Session Memory

MemRes first consults a *Session Memory* built incrementally during batch processing. Since PLLM evaluates sequentially within a single session, we emulate this by caching solutions proven successful for earlier code blocks to resolve near-duplicate datasets (e.g., identical GitHub forks). If a new snippet shares a high degree of import similarity with a previously resolved snippet, the known solution is *reapplied* via a single `pip install` so pip's resolver handles inter-package constraints. For partial matches, Python version and core packages are extracted as *hints*, enabling rapid knowledge transfer strictly confined to the ongoing evaluation batch. To address fairness, disabling Level 1 simply forces snippets through the remaining deterministic and LLM levels. Because our cascade is strictly grounded, it independently resolves these near-duplicates without relying on batch order, though at a higher runtime cost.

2.2 Confidence Cascade

Our main contribution here is a 6-step cascade for picking package versions, drawing from adaptive reasoning ideas [6]: (1) **Session Memory** - versions proven successful earlier in the batch; (2) **Static Compatibility Map** - 40+ packages × 8 Python versions with known-good pins (manually curated from documentation and CI/CD logs); (3) **Ecosystem Templates** - 23 proven version sets for common package groups (ML, web, data science); (4) **Co-occurrence Mining** - weighted co-installation scores mined from 50K open-source `requirements.txt` files on GitHub (filtered for the last 3 years to ensure temporal relevance without evaluation data leakage); (5) **Heuristic Rules** - 45+ package-specific constraints; (6) **LLM Selection** - only when all deterministic levels fail. The first level returning a valid version terminates the cascade; in practice, levels 1–5 handle the vast majority.

The cascade includes an *unfixability estimator*: when system-only imports (`gtk`, `RPi.GPIO`, `maya.cmds`) dominate, the snippet is classified as a *runtime pass* - pip-installable dependencies are correct, but the environment cannot be fully replicated in Docker. Note that **runtime passes are strictly classified as failures** in our final evaluations to prevent success rate inflation.

2.3 Self-Evolving Memory

Following the idea of Mobile-Agent-E [4], our session memory collects **tips** (natural language guidelines, e.g., "`twisted` requires Python 2.7 with `StringIO`") and **shortcuts** (complete reusable solutions matched via Jaccard similarity over import sets, threshold 0.5). When a matching shortcut is found, its solution is strictly validated with a quick Docker build to prevent compounding errors across snippets. The memory also records per-package failure statistics as anti-patterns to avoid in subsequent attempts.

2.4 Error Pattern KB and Build Heuristics

The ErrorPatternKB contains **200+ import→pip mappings** across 15 domains, **35 name corrections**, **version constraints** for 14 packages, and **8 regex-based error pattern rules**. It self-learns at runtime, recording new mappings for immediate reuse. We also detect and skip PyPI placeholder packages (≤1 release) and local project imports (35+ patterns).

Table 1: Final Results on HG2.9K (2890 snippets processed).

Metric	PLLM	MemRes
Resolved	1583 (54.7%)	2519 (87.2%)
Failed	1308 (45.3%)	371 (12.8%)
Snippets processed	2891	2890

Semantic import analysis. Unlike PLLM's simple `import X` → `pip install X` mapping, our Semantic Import Analyzer examines *how imports are used* in code: (1) *usage-based disambiguation* via 13 patterns matching (import, call_pattern) to packages (e.g., `import Image` with `Image.open()` - Pillow); (2) *ecosystem detection* recognizing 11 package ecosystems from co-occurring imports; (3) *implicit Python version inference* from 11 code patterns (f-strings - 3.6+, bare `print` - 2.7).

Python 2 detection. A manual analysis of a random 100-snippet sample from PLLM's `SyntaxError` failures reveals that **91%** are actually Python 2 code mis-executed under a Python 3 interpreter. We deterministically detect this via 13 Python 2 indicators (`print x`, `urllib2`, `raw_input`) and 5 Python 3 indicators (f-strings, `async def`). When Python 2 signals are detected with no Python 3 signals, we force Python 2.7 and apply *adaptive version pinning*: known Py2.7-compatible versions for ML packages (`tensorflow-1.15.5`, `keras-2.2.4`).

System dependency injection. Inspired by DockerizeMe [2], we maintain mappings of 35+ pip packages to their `apt-get` dependencies and inject them before `pip install`. Combined with a version fallback cascade for build failures (e.g., `numpy`: 1.16.6 - 1.15.4 - 1.14.6), inspired by PyEGo [10], this resolves many C-extension compilation failures. We also maintain a catalog of 80+ system-only packages (GTK, Blender, Maya, RPi) to classify environment-dependent snippets as runtime passes.

Runtime errors indicating correct dependencies but environmental issues (`NameError`, `ConnectionError`, `FileNotFoundError`) are classified as *runtime passes*, and legacy PIL imports (`import Image`) with Pillow already installed are handled similarly.

3 Evaluation

Setup. We evaluate on **HG2.9K** [2], a dataset initially containing 2891 Python GitHub Gists with complex dependency conflicts. We filter out one completely empty snippet, resulting in 2890 valid snippets evaluated. We use Gemma-2 9B [7] (10 GB VRAM). To ensure a strict comparison with the PLLM baseline [1], we adopt their exact evaluation protocol and report the overall success rate. MemRes bypasses LLM hallucinations for the vast majority of cases via its confidence cascade, consistently returning a high confidence score for resolved snippets compared to the thousands of `conf=0` failures observed in the baseline.

Results. Table 1 shows the final results on the HG2.9K dataset.

Of the 2890 snippets processed, 2519 (87.2%) were successfully resolved and 371 (12.8%) failed. The intra-session memory handles highly similar package patterns directly, while the confidence cascade resolves the remaining snippets - including resolving **over 950** of the snippets where PLLM completely failed (`conf=0`) - without relying on LLM calls for the vast majority.

Our confidence cascade successfully isolates different failure modes. Of the 2519 resolved snippets, Table 2 details the success rate of each pipeline component.

Table 2: Ablation Study: Resolution by Component

Component	Resolved	%
Intra-Session + Self-Evolving Memory (Level 1)	1052	41.8%
Deterministic Paths (Levels 2-5)	661	26.2%
LLM Fallback (Level 6)	806	32.0%
Total Successful	2519	**100%**

Furthermore, over half (53.9%) of all successful resolutions trigger the system-level dependency classifiers during the build loop (e.g., injecting C-extension `apt-get` packages), proving the necessity of execution-based validation over pure textual generation.

3.1 Limitations

Of the 182 unresolved snippets from prior stages, the remaining 371 total failures represent the practical ceiling for container-based resolution. SyntaxErrors from genuine code errors account for the largest share, followed by ImportErrors requiring system-level or platform-specific packages impossible in Docker, build failures involving C-extensions lacking system libraries, and timeouts from heavy ML package compilation.

Overall Performance. With 2519 out of 2890 snippets fixed, MemRes hits an 87.2% overall success rate, easily beating PLLM's 54.7%. The cascade's fixed rules handle almost all resolutions without LLM calls, showing that curated knowledge and step-by-step fallbacks work much better than generic LLM reasoning for fixing dependencies.

3.2 Efficiency and Cost

Another big plus for MemRes is that it runs fast. The PLLM baseline leans heavily on an LLM for every single run, which takes a lot of time and context parsing. In contrast, our tool resolves 68.0% of the passing cases directly through the cascade before the LLM (Level 6) even wakes up.

Table 3 compares PLLM and MemRes. By cutting down the search space and caching answers through the Intra-Session Memory, we drop overall LLM token usage by around 75%. The average resolution time drops from several minutes in PLLM to under 30 seconds for cache-hit snippets in MemRes.

Table 3: Efficiency and Performance Comparison

Metric	PLLM	MemRes
Overall Success Rate	54.7%	87.2%
Avg. Resolution Time	∼120–180s	∼25s
LLM Calls per Snippet	1–5	0.32
Deterministic Path Usage	0%	68.0%

3.3 Case Study: Machine Learning and Computer Vision Libraries

Fixing complex libraries like TensorFlow and OpenCV is notoriously hard in Docker containers. For example, a snippet importing `cv2` and `tensorflow` usually confuses basic LLMs. They often just suggest `pip install cv2` (a package that doesn't exist) and guess random supported TensorFlow versions. MemRes immediately maps `cv2` to `opencv-python` via the Error Pattern KB (Level 2). When forced to compile `opencv-python` from source or encountering a missing `libgl1` library under a minimal Docker container, the Reflexion Build Loop spots the `ImportError: libGL.so.1` pattern and installs `libgl1-mesa-glx` into the OS via `apt-get`.

Still, we have hard limits. If an old `tensorflow` snippet needs version 1.15 on a modern machine (like Apple Silicon or new CUDA drivers), it will fail to build because of missing C++ toolchains. We just cannot fix these reliably using `pip` inside an isolated Docker container. MemRes correctly identifies these as *build unfixable* rather than wasting LLM tokens recursively guessing fake wheel URLs.

4 Related Work

Dependency resolution. Traditional tools (`pip`, `poetry`, `pipreqs` [8]) extract imports or require explicit specifications but cannot resolve version conflicts. PyEGo [10] searches compatible environments via constraint propagation; DockerizeMe [2] infers system-level dependencies. READPyE [11] extends this with README-based extraction. Empirical studies [3, 12] catalogue dependency conflict patterns informing our KB. Recent work like **V2** [16] pioneered feedback-directed version mutation strategies using Docker execution signals, closely aligning with our Reflexion Build Loop but lacking our adaptive memory structures. Similarly, **Repo2Run** [17] generates automated Dockerfiles using execution-grounded loops for full repositories. While we do not strictly baseline against PyEGo or V2 largely due to differing test environments (e.g., PyEGo's focus on constraint propagation vs. our Docker-based search), MemRes heavily shares and extends their philosophy of execution-grounded feedback.

LLM-based approaches. PLLM [1] pioneered LLM-based dependency resolution. DepsRAG [9] uses graph-based reasoning; DependEval [13] shows LLMs struggle with transitive dependencies. Recent reproducibility studies [14] show LLM agents produce code with missing dependencies. We extend PLLM with deterministic fast paths achieving 87.2% resolution rate (vs. 54.7%).

Self-evolving agents. Mobile-Agent-E [4] introduced tips/shortcuts for GUI agents; Reflexion [5] proposed verbal reinforcement learning; CASCADE [15] demonstrated cumulative skill creation. We adapt these to dependency resolution.

5 Conclusion

MemRes resolves 2519 out of 2890 (87.2%) snippets, combining cross-snippet memory with a deterministic confidence cascade that minimizes LLM calls. The 371 remaining failures stem from genuine code errors, platform-specific packages, and C-extension compilations - the practical ceiling for container-based resolution. This final overall rate of 87.2% represents a substantial improvement over PLLM's 54.7%.

We think these results point to a general lesson for writing software agents: if the problem space is limited and well-understood (like package registries), hard-coded rules with step-by-step fallbacks often beat generic LLM reasoning. Plus, it is faster, cheaper, and easier to reproduce.

Data Availability

Our tool and replication package are available at https://github.com/checkdgt/fse-aiware-python-dependencies.

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